

Table 1: USER (Strong entity)

Central login account for every system user. This replaces the earlier USER name for clearer ER wording.

Field name	Data type	Key	Null	Description
user_id	INT	PK&AI	No	Unique user account ID.
role_name	VARCHAR(50)	FK	No	Role name such as System Administrator, CEO, Driver or Finance Officer.
username	VARCHAR(50)	Unique	No	Login username.
password_hash	VARCHAR(255)	-	No	Encrypted/hashed password value.
email	VARCHAR(100)	Unique	No	User email address.
is_active	TINYINT(1)	-	No	Shows whether account is active.
created_at	DATETIME	-	No	Account creation date and time.

Table 2: EMPLOYEE (Strong entity)

Stores staff details for every employee, including the System Administrator.

Field name	Data type	Key	Null	Description
employee_id	INT	PK&AI	No	Unique employee ID.
user_id	INT	FK, Unique	Yes	Links employee to USER_ACCOUNT if the employee can log in.
full_name	VARCHAR(100)	-	No	Employee full name.
nic	VARCHAR(20)	Unique	No	National identity card number.
contact_no	VARCHAR(20)	-	No	Employee contact number.
address	TEXT	-	Yes	Employee address.
job_title	VARCHAR(100)	-	No	Employee job title or position.
hire_date	DATE	-	No	Date employee joined the business.
base_salary	DECIMAL(10,2)	-	No	Basic monthly salary.
employment_status	VARCHAR(30)	-	No	Active, inactive, resigned or terminated status.

Table 3: LOGIN_HISTORY (Weak entity)

Tracks login and logout activity for security monitoring.

Field name	Data type	Key	Null	Description
login_id	INT	PK&AI	No	Unique login history record ID.
user_id	INT	FK	No	User account that attempted login.
login_time	DATETIME	-	No	Login attempt date and time.
logout_time	DATETIME	-	Yes	Logout date and time.

Table 4: AUDIT_LOG (Weak entity)

Stores user activity logs and supports audit reports.

Field name	Data type	Key	Null	Description
log_id	INT	PK&AI	No	Unique audit log ID.
user_id	INT	FK	No	User account that performed the action.
action	VARCHAR(100)	-	No	Action performed by the user.
target_table	VARCHAR(100)	-	Yes	Table/module affected by the action.
target_id	INT	-	Yes	Record ID affected by the action.
timestamp	DATETIME	-	No	Date and time of the action.

Table 5: CUSTOMER (strong)

Stores customer details and current loyalty point balance.

Field name	Data type	Key	Null	Description
customer_id	INT	PK&AI	No	Unique customer ID.
name	VARCHAR(100)	-	No	Customer full name or business name.
contact_no	VARCHAR(20)	-	No	Customer phone number.

Field name	Data type	Key	Null	Description
email	VARCHAR(100)	Unique	Yes	Customer email address.
address	TEXT	-	Yes	Customer address.
loyalty_points	INT	-	No	Current loyalty point balance calculated from purchases.

Table 6: COMPLAINT (weak)

Stores customer complaints and escalations.

Field name	Data type	Key	Null	Description
complaint_id	INT	PK&AI	No	Unique complaint ID.
customer_id	INT	FK	No	Customer who raised the complaint.
officer_id	INT	FK	Yes	Employee/CRO handling the complaint.
description	TEXT	-	No	Complaint details.
status	VARCHAR(30)	-	No	Open, in progress, resolved or escalated status.
created_date	DATETIME	-	No	Complaint created date and time.
resolved_date	DATETIME	-	Yes	Date and time complaint was resolved.
escalated_to	INT	FK	Yes	Manager/CEO employee ID if complaint is escalated.

Table 7: PRODUCT (strong)

Main product catalogue used by sales and inventory.

Field name	Data type	Key	Null	Description
product_id	INT	PK&AI	No	Unique product ID.

Field name	Data type	Key	Null	Description
product_name	VARCHAR(100)	-	No	Product name.
selling_price	DECIMAL(10,2)	-	No	Current selling price.
min_stock_level	INT	-	No	Minimum stock level for low-stock alerts.

Table 8: SUPPLIER (strong)

Stores supplier details for purchases and payments.

Field name	Data type	Key	Null	Description
supplier_id	INT	PK&AI	No	Unique supplier ID.
supplier_name	VARCHAR(100)	-	No	Supplier company/name.
contact_no	VARCHAR(20)	-	No	Supplier phone number.
email	VARCHAR(100)	-	Yes	Supplier email address.
address	TEXT	-	Yes	Supplier address.
payment_terms	VARCHAR(100)	-	Yes	Credit period or payment condition.

Table 9: STOCK_BATCH (weak)

Tracks incoming stock batches, expiry and damaged/expired stock.

Field name	Data type	Key	Null	Description
batch_id	INT	PK&AI	No	Unique stock batch ID.
product_id	INT	FK	No	Product stored in the batch.
supplier_id	INT	FK	Yes	Supplier who supplied the batch.
received_date	DATE	-	No	Date stock batch was received.

Field name	Data type	Key	Null	Description
expiry_date	DATE	-	Yes	Expiry date, if applicable.
original_quantity	INT	-	No	Quantity received in the batch.
current_quantity	INT	-	No	Current remaining quantity in the batch.
status	VARCHAR(30)	-	No	Available, damaged, expired or depleted status.

Table 10: STOCK_MOVEMENT (weak)

Records stock in, sales deduction, returns, damage, expiry and adjustments.

Field name	Data type	Key	Null	Description
movement_id	INT	PK&AI	No	Unique stock movement ID.
product_id	INT	FK	No	Product whose stock changed.
batch_id	INT	FK	Yes	Related stock batch if batch-level tracking is used.
movement_type	VARCHAR(30)	-	No	Stock-in, sale, return, damage, expiry or adjustment.
quantity	INT	-	No	Quantity moved. Use positive value with movement_type.
movement_date	DATETIME	-	No	Date and time of movement.
recorded_by	INT	FK	No	Employee who recorded the movement.
notes	TEXT	-	Yes	Extra movement notes.

Table 11: SALES_ORDER (strong entity)

Stores customer orders and order status. Discount rules are simplified to a single discount_amount field.

Field name	Data type	Key	Null	Description
order_id	INT	PK&AI	No	Unique sales order ID.
customer_id	INT	FK	No	Customer who placed the order.
officer_id	INT	FK	No	Employee who processed the order.

Field name	Data type	Key	Null	Description
order_date	DATETIME	-	No	Date and time order was created.
status	VARCHAR(30)	-	No	Pending, confirmed, delivered, cancelled or returned status.
discount_amount	DECIMAL(10,2)	-	No	Discount amount manually applied to the order.
tax_amount	DECIMAL(10,2)	-	No	Tax amount for the order.
total_amount	DECIMAL(10,2)	-	No	Final order amount.
is_credit	TINYINT(1)	-	No	Shows whether order is a credit order.
credit_approved_by	INT	FK	Yes	Employee/manager who approved credit order.

Table 12: ORDER_ITEM (weak)

Stores products inside a sales order.

Field name	Data type	Key	Null	Description
item_id	INT	PK&AI	No	Unique order item ID.
order_id	INT	FK	No	Sales order ID.
product_id	INT	FK	No	Ordered product ID.
quantity	INT	-	No	Quantity ordered.
unit_price	DECIMAL(10,2)	-	No	Selling price at the time of order.
line_total	DECIMAL(10,2)	-	No	Quantity multiplied by unit price.

Table 13: INVOICE (strong)

Stores invoice details generated from confirmed orders.

Field name	Data type	Key	Null	Description
invoice_id	INT	PK&AI	No	Unique invoice ID.
order_id	INT	FK, Unique	No	Confirmed sales order linked to the invoice.

Field name	Data type	Key	Null	Description
issue_date	DATE	-	No	Invoice issue date.
subtotal	DECIMAL(10,2)	-	No	Invoice subtotal before discount and tax.
discount_amount	DECIMAL(10,2)	-	No	Discount amount shown on invoice.
tax_amount	DECIMAL(10,2)	-	No	Tax amount shown on invoice.
total_amount	DECIMAL(10,2)	-	No	Final invoice total.
payment_status	VARCHAR(30)	-	No	Unpaid, partially paid, paid or overdue status.

issue

Table 14: PAYMENT (strong)

Tracks customer payments and outstanding receivables.

Field name	Data type	Key	Null	Description
payment_id	INT	PK&AI	No	Unique customer payment ID.
invoice_id	INT	FK	No	Invoice being paid.
customer_id	INT	FK	No	Customer who made the payment.
received_by	INT	FK	No	Employee who received/recorded payment.
amount	DECIMAL(10,2)	-	No	Payment amount.
payment_method	VARCHAR(30)	-	No	Cash, card, bank transfer or cheque method.
payment_date	DATETIME	-	No	Date and time payment was received.
payment_status	VARCHAR(30)	-	No	Completed, pending, failed or refunded status.

Table 15: VEHICLE (strong entity)

Stores fleet vehicle details.

Field name	Data type	Key	Null	Description
vehicle_id	INT	PK&AI	No	Unique vehicle ID.

Field name	Data type	Key	Null	Description
plate_number	VARCHAR(30)	Unique	No	Vehicle registration plate number.
vehicle_type	VARCHAR(50)	-	No	Van, lorry, bike or other vehicle type.
capacity	VARCHAR(50)	-	Yes	Vehicle carrying capacity.

Table 16: DELIVERY (weak entity)

Stores delivery plan, driver, vehicle and progress.

Field name	Data type	Key	Null	Description
delivery_id	INT	PK&AI	No	Unique delivery ID.
order_id	INT	FK, Unique	No	Sales order fulfilled by this delivery.
driver_id	INT	FK	No	Employee assigned as driver.
vehicle_id	INT	FK	No	Vehicle used for the delivery.
scheduled_date	DATE	-	No	Planned delivery date.
route_details	TEXT	-	Yes	Delivery route or address details.
status	VARCHAR(30)	-	No	Scheduled, in transit, delivered, delayed or cancelled status.
transport_cost	DECIMAL(10,2)	-	No	Delivery transport cost.

Table 17: DELIVERY_PROOF (weak entity)

Stores proof of delivery uploaded by driver.

Field name	Data type	Key	Null	Description
proof_id	INT	PK&AI	No	Unique delivery proof ID.
delivery_id	INT	FK, Unique	No	Delivery record confirmed by this proof.
image_url	VARCHAR(255)	-	Yes	Path/URL of proof image.

Field name	Data type	Key	Null	Description
uploaded_at	DATETIME	-	No	Date and time proof was uploaded.
received_by_name	VARCHAR(100)	-	No	Name of customer/person who received goods.
notes	TEXT	-	Yes	Extra proof notes.

Table 18: DELIVERY_ISSUE (weak)

Stores delivery problems and delayed delivery reasons.

Field name	Data type	Key	Null	Description
issue_id	INT	PK&AI	No	Unique delivery issue ID.
delivery_id	INT	FK	No	Delivery with the issue.
reported_by	INT	FK	No	Employee/driver who reported issue.
issue_description	TEXT	-	No	Details of delivery issue.
issue_date	DATETIME	-	No	Date and time issue was reported.

Table 19: ATTENDANCE (weak)

Stores daily attendance and overtime information.

Field name	Data type	Key	Null	Description
attendance_id	INT	PK&AI	No	Unique attendance record ID.
employee_id	INT	FK	No	Employee whose attendance is recorded.
recorded_by	INT	FK	Yes	Employee/admin who recorded attendance.
date	DATE	-	No	Attendance date.
clock_in	TIME	-	Yes	Clock-in time.
clock_out	TIME	-	Yes	Clock-out time.

Field name	Data type	Key	Null	Description
overtime_hours	DECIMAL(5,2)	-	No	Overtime hours worked.

Table 20: LEAVE_REQUEST (weak)

Stores employee leave requests and approvals.

Field name	Data type	Key	Null	Description
leave_id	INT	PK&AI	No	Unique leave request ID.
employee_id	INT	FK	No	Employee requesting leave.
approved_by	INT	FK	Yes	Employee/manager who approved or rejected leave.
start_date	DATE	-	No	Leave start date.
end_date	DATE	-	No	Leave end date.
reason	TEXT	-	Yes	Reason for leave request.
status	VARCHAR(30)	-	No	Pending, approved or rejected status.
requested_date	DATETIME	-	No	Date and time leave was requested.

Table 21: PAYROLL (weak)

Stores salary calculation for a pay period.

Field name	Data type	Key	Null	Description
payroll_id	INT	PK&AI	No	Unique payroll record ID.
employee_id	INT	FK	No	Employee receiving salary.
generated_by	INT	FK	No	Employee/admin who generated payroll.
period	VARCHAR(20)	-	No	Salary period such as 2026-06.
base_salary	DECIMAL(10,2)	-	No	Basic salary for the period.

Field name	Data type	Key	Null	Description
overtime_pay	DECIMAL(10,2)	-	No	Overtime payment amount.
deductions	DECIMAL(10,2)	-	No	Salary deductions.
net_pay	DECIMAL(10,2)	-	No	Final salary after additions and deductions.
generated_date	DATETIME	-	No	Date and time payroll was generated.

Table 22: PURCHASE_ORDER (weak)

Stores supplier purchase requests and major purchase approval.

Field name	Data type	Key	Null	Description
purchase_id	INT	PK&AI	No	Unique purchase order ID.
supplier_id	INT	FK	No	Supplier receiving purchase order.
requested_by	INT	FK	No	Employee who requested purchase.
approved_by	INT	FK	Yes	Manager/CEO who approved purchase.
request_date	DATE	-	No	Date purchase was requested.
approval_status	VARCHAR(30)	-	No	Pending, approved or rejected status.
total_amount	DECIMAL(10,2)	-	No	Total purchase amount.
expected_date	DATE	-	Yes	Expected stock receiving date

Table 23: PURCHASE_ITEM (weak)

Stores products included in a purchase order.

Field name	Data type	Key	Null	Description
purchase_item_id	INT	PK&AI	No	Unique purchase item ID.
purchase_id	INT	FK	No	Purchase order ID.
product_id	INT	FK	No	Product being purchased.

Field name	Data type	Key	Null	Description
quantity	INT	-	No	Quantity purchased.
unit_cost	DECIMAL(10,2)	-	No	Buying cost per unit.
line_total	DECIMAL(10,2)	-	No	Quantity multiplied by unit cost.

Table 24: SUPPLIER_PAYMENT (weak)

Stores supplier payment records.

Field name	Data type	Key	Null	Description
supplier_payment_id	INT	PK&AI	No	Unique supplier payment ID.
supplier_id	INT	FK	No	Supplier receiving payment.
purchase_id	INT	FK	Yes	Related purchase order if payment settles a purchase.
paid_by	INT	FK	No	Employee who made/recorded payment.
amount	DECIMAL(10,2)	-	No	Payment amount.
payment_date	DATETIME	-	No	Date and time supplier payment was made.
status	VARCHAR(30)	-	No	Paid, pending, partial or cancelled status.

Table 25: BUDGET_PLAN (strong)

Stores budgets and financial plans without department entities.

Field name	Data type	Key	Null	Description
budget_id	INT	PK&AI	No	Unique budget plan ID.
budget_purpose	VARCHAR(150)	-	No	Purpose/category of the budget plan.
period	VARCHAR(20)	-	No	Budget period such as 2026-Q1 or 2026.
amount	DECIMAL(12,2)	-	No	Budget amount.
prepared_by	INT	FK	No	Employee who prepared budget plan.

Field name	Data type	Key	Null	Description
approved_by	INT	FK	Yes	CEO/head manager who approved budget.
status	VARCHAR(30)	-	No	Draft, pending, approved or rejected status.

Table 26: FINANCIAL_RECORD (strong)

Stores income, expense, tax, reconciliation and finance records.

Field name	Data type	Key	Null	Description
record_id	INT	PK&AI	No	Unique financial record ID.
recorded_by	INT	FK	No	Employee/finance officer who recorded entry.
related_payment_id	INT	FK	Yes	Customer payment linked to this record.
supplier_payment_id	INT	FK	Yes	Supplier payment linked to this record.
type	VARCHAR(30)	-	No	Income, expense, tax, adjustment or reconciliation type.
amount	DECIMAL(12,2)	-	No	Financial record amount.
description	TEXT	-	Yes	Financial record description.
record_date	DATE	-	No	Date of financial record.
tax_amount	DECIMAL(10,2)	-	No	Tax amount related to this record.

Table 27: ANNOUNCEMENT (Strong entity)

Stores internal announcements sent by CEO/head manager.

Field name	Data type	Key	Null	Description
announcement_id	INT	PK&AI	No	Unique announcement ID.
sent_by	INT	FK	No	Employee who sent the announcement.
title	VARCHAR(150)	-	No	Announcement title.

Field name	Data type	Key	Null	Description
message	TEXT	-	No	Announcement message.
receiver_id	INT	FK	Yes	Employee who receives the announcement.
created_at	DATETIME	-	No	Date and time announcement was created.

Table 28: BACKUP_RECORD (Strong entity)

Field name	Data type	Key	Null	Description
Backup_id	INT	PK&AI	No	Unique backup/restore record ID
Backup_type	VARCHAR(30)	-	No	Type of action: Backup or Restore
File_path	VARCHAR(225)	-	Yes	Backup file name or file location
date	DATETIME	-	Yes	Date and time action was performed
note	TEXT	-	Yes	Extra notes about the backup/restore